Youngjin (Eric) Noh — CV

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Research interests: contact and friction modeling of continuum and soft robots; soft-robot control; robot manipulation and learning from demonstration.

Education

Ulsan National Institute of Science and Technology (UNIST), Ulsan, Korea
B.S. in Mechanical Engineering · Mar 2022 – Feb 2029 (expected) · GPA 3.86 / 4.3 (major 4.03 / 4.3)
Leave of absence for military service, Jun 2023 – Dec 2024

The Hong Kong University of Science and Technology (HKUST), Hong Kong
Exchange student, School of Engineering · Feb – May 2027 (incoming)

Relevant coursework: Applied Linear Algebra, Differential Equations, Signals and Systems, Dynamics, Classical Mechanics I, Solid Mechanics I–II, Electrical Engineering Programming, AI-Based Digital Manufacturing. **In progress (Fall 2026):** Introduction to Robotics, Dynamic Systems and Control, Numerical Analysis, Optimization Theory.

Research Experience

Intelligent Robotics Lab (IntRo), UNIST — Undergraduate Researcher · Dec 2025 – present
Advisor: Prof. Junhyoung Ha

Law-preserving resolution selection for contact-accurate continuum-robot simulation (project lead)

- Showed in pilot runs that refining a MuJoCo rigid-chain model of a tendon-driven continuum robot can change its contact law: per-length contact stiffness rose by about 95–108% from 10 to 80 links once link rotational inertia mattered.
- Developed a law-preserving parameterization that scales each contact row by the engine's inverse-inertia estimate; it held the target per-row stiffness to machine precision in the same pilots.
- Designing an output-oriented resolution-selection method (adjoint-based estimates of wrench and squeeze-force changes) checked against an independent rod-model reference, toward a simulation paper.

Teleoperation feedback for imitation-learning data collection (UNIST UIRP team project, May 2026 – Sep 2026)

- Leader–follower teleoperation on OpenManipulator-X (ROS 2 Jazzy, ros2_control, Dynamixel), comparing six feedback conditions — from unilateral control to variable stiffness with tactile feedback — as data-collection methods for imitation learning.
- Implemented MuJoCo-based gripper force feedback from simulated contact forces; set up the environment and data pipeline, ran calibration, and improved the tactile tokenizer.

- Trained manipulation policies on frozen DINOv2 visual features with action chunking and evaluated them closed-loop in MuJoCo.

Selected Course Projects

- **Casting-defect image classification** — AI-based Digital Manufacturing, Spring 2026. ResNet-style CNN, 98.44% validation accuracy.
- **Battery thermal-runaway detection** (team) — AI-based Digital Manufacturing, Spring 2026. Progressed from a 1D-CNN (76.6%) to semi-supervised autoencoder pretraining (88.5%), then multimodal and image-only ResNet-50 models on an AIHub dataset.
- **PINN for inverse thermal-conductivity estimation** — Heat Transfer (MEN310). Diagnosed loss imbalance, non-dimensionalized the problem and trained with Adam followed by L-BFGS; assessed when a PINN adds value over FDM-based inverse optimization.

Skills

- **Robotics & simulation:** ROS 2 (Jazzy), ros2_control, Dynamixel SDK, MuJoCo (rigid-chain and cable models, contact and constraint tuning)
- **Machine learning:** PyTorch, LeRobot / ACT, DINOv2 feature pipelines, CNNs (ResNet), physics-informed neural networks
- **Programming:** Python, C++, MATLAB
- **Tools:** Linux (Ubuntu), Git

Other

- **Military service:** Republic of Korea, Jun 2023 – Dec 2024 (completed)
- **Languages:** Korean (native), English (TOEIC 825; all UNIST courses are taught in English)